# E-COMMERCE SALES DASHBOARD

PROJECT REPORT

Data Cleaning • Exploratory Data Analysis • Business Insights

Tools: Microsoft Excel, Python, Tableau / BI-style Visualizations

Prepared as a portfolio-ready project report based on the provided e-commerce workbook and visualization screenshots.

| | |
|---|---|
| **Project Type** | Sales and order analytics |
| **Data Period Visible** | January 2024 – June 2026 (monthly chart) |
| **Dataset Structure** | Order, customer, product, geography, payment, shipping, status, price, cost and sales fields |
| **Prepared on** | 12 October 2026 |

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Course:-CSE(AI&DS)

Aspiring as:-Data Analyst

# 1. Executive Summary

This project analyzes e-commerce orders to understand sales performance, profitability, customer activity, product-category contribution, order status, payment preferences, shipping choices and geographic distribution. The workflow begins with a raw Excel sheet, uses a cleaned-data sheet for consistent analysis, and presents findings through charts and dashboard-style visualizations.

The supplied overview dashboard displays the following headline figures. These values are transcribed from the screenshot and should be revalidated against the final workbook before formal submission.

| Metric | Value shown | Meaning |
|---|---|---|
| Total Orders | 1,018 | Order volume represented in the overview |
| Total Sales | 13,01,226.80 | Sum of sales shown on the dashboard |
| Total Profit | 4,19,243.69 | Sum of profit shown on the dashboard |
| Total Customers | 1,018 | Customer count shown; confirm whether IDs are unique |
| Average Order Value / Average Orders | 1,278.22 | Label in screenshot says “Average Orders”; confirm calculation definition |

**Important reporting note: The individual Python charts are specifically titled “Delivered Orders”, while the overview dashboard appears to aggregate all order statuses. Therefore, totals from the overview should not be directly compared with delivered-only charts unless the same status filter is applied.**

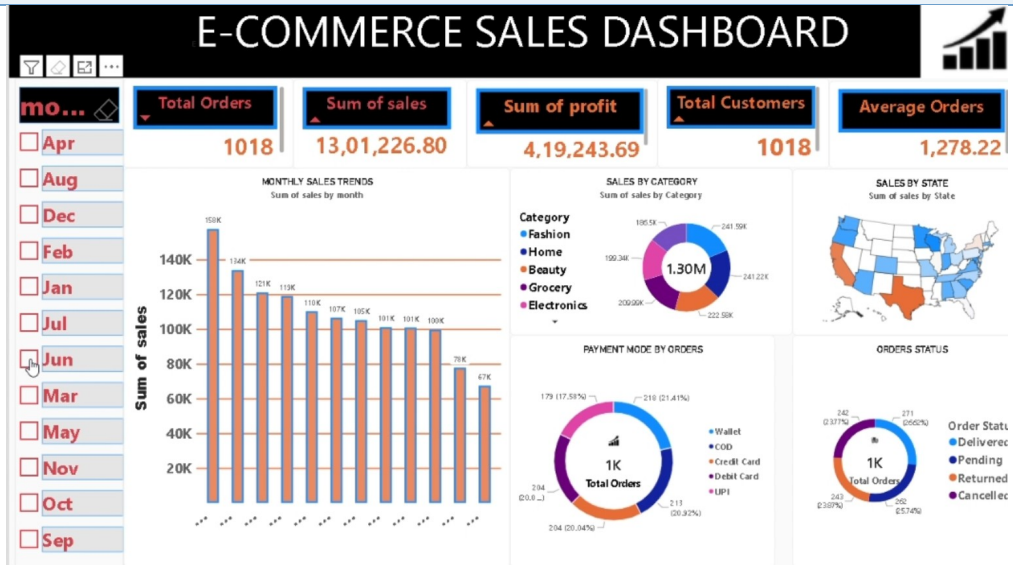



*Figure 1. Overview dashboard screenshot supplied for the project.*

# 2. Objectives and Scope

• Prepare a consistent dataset suitable for analysis and dashboarding.

• Measure overall sales, profit and order activity.

• Identify monthly sales changes and category-level performance.

• Understand order statuses, payment modes and shipping choices.

• Review customer-level and state-level sales patterns.

• Translate chart observations into practical business recommendations.

# 3. Dataset Description

The workbook screenshots show two worksheets: “raw data” and “clean data”. The raw worksheet contains transaction-level fields, while the cleaned worksheet includes standardized fields and derived values for analysis.

Fields visible in the raw/clean data screenshots include:

| Group | Fields visible |
|---|---|
| Order and dates | Order ID, Order Date, Ship Date, Delivery Days, Month, Quarter, Year |
| Customer and location | Customer ID, Customer Name, State, City, Gender, Age |
| Product details | Product Name, Category, Sub Category, Product ID, Brand, Ratings |
| Order fulfillment | Shipping Mode, Payment Mode, Order Status |
| Financial measures | Quantity, Discount %, Unit Price, Cost, Sales, Profit |

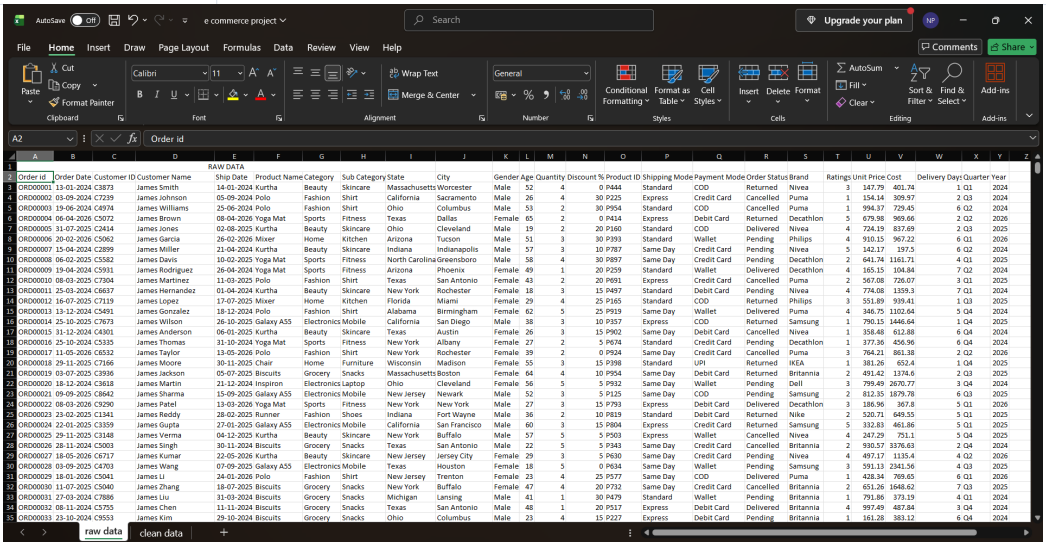



*Figure 2. Raw data worksheet screenshot.*

# 4. Data Preparation and Cleaning

The cleaning stage is intended to make the dataset accurate, consistent and ready for analysis. The report reflects the two-sheet workflow visible in the workbook; the exact operations should be documented according to the checks actually performed.

**Structure check:** Confirm column headers, row boundaries and that each row represents an order or order line consistently.

**Duplicate check:** Check duplicate records and investigate repeated Order IDs before deleting anything, because one order can sometimes contain multiple line items.

**Missing values:** Identify blanks in important fields such as dates, category, customer, quantity, price, cost and status; document the treatment used.

**Standardization:** Normalize text values for category, payment mode, shipping mode, state and order status so spelling and capitalization do not split the same group.

**Data types:** Convert order/ship dates to date values; set quantity, age, discount, rating, unit price, cost, sales and profit to suitable numeric types.

**Derived fields:** Use a consistent date source to derive Month, Quarter and Year. Confirm that calculated Sales and Profit formulas match the business definitions.

**Validation:** Check impossible values, such as negative quantities or invalid ratings, and reconcile totals between the cleaned data and dashboard.

**A reliable cleaning process should preserve the original raw worksheet and make transformations reproducible. The clean sheet should be treated as the analysis source, while the raw sheet remains the audit/reference copy.**

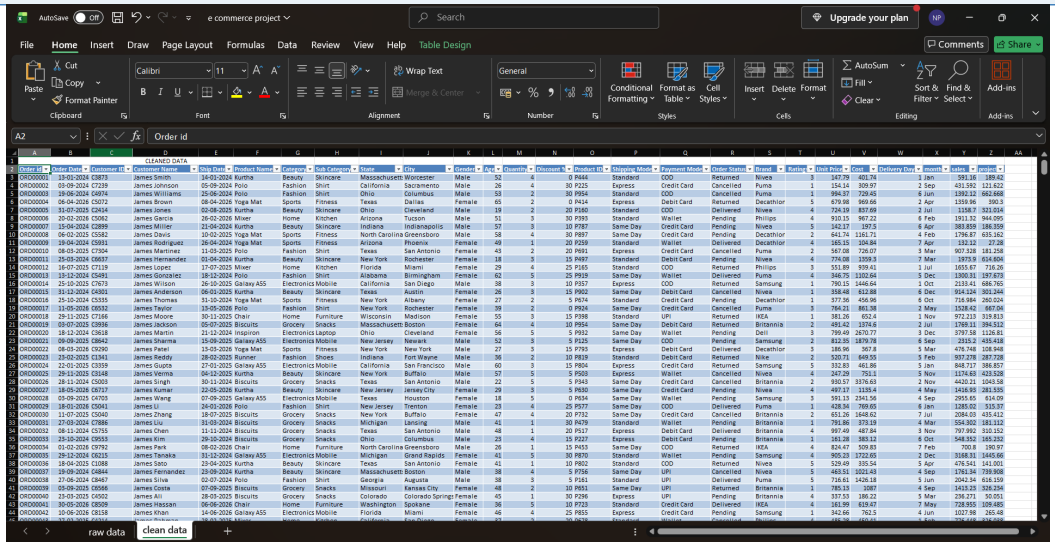



*Figure 3. Clean data worksheet screenshot.*

# 5. Analysis Methodology

The analysis is organized into five layers:

| Layer | Method | Output |
|---|---|---|
| 1. Summary KPIs | Aggregate order count, sales and profit; verify distinct customers and the average-order metric. | Headline performance indicators |
| 2. Time analysis | Group sales by month and sort months chronologically. | Monthly sales trend chart |
| 3. Product analysis | Group delivered-order sales by category. | Category contribution chart |
| 4. Operations analysis | Count orders by status, payment mode and shipping mode. | Operational mix charts |
| 5. Customer and geography | Rank customer sales and aggregate sales by state. | Top-customer chart and state map |

# 6. Dashboard and Chart Findings

## 6.1 Monthly sales trend

The delivered-orders monthly chart spans January 2024 through June 2026. Sales fluctuate rather than following a steady upward trend. The chart has visible peaks around August 2024 and April 2025, and lower values around October 2024, February 2026 and June 2026. Because the final month may be incomplete, June 2026 should not be compared with full months without checking the reporting cutoff.

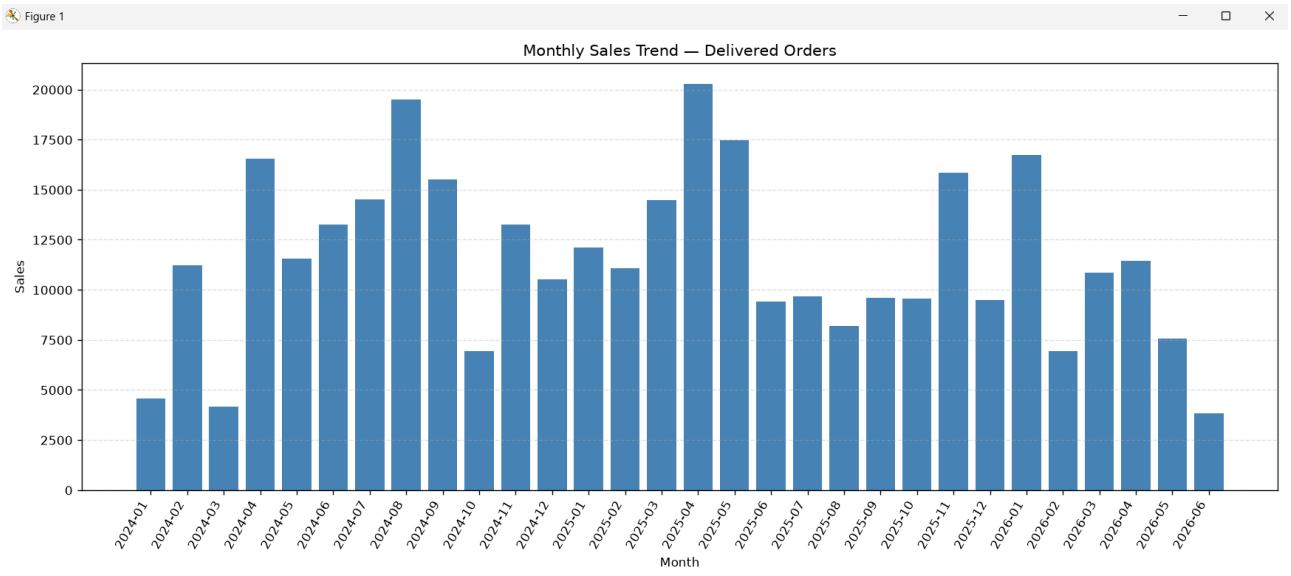



*Figure 4. Monthly sales trend for delivered orders.*

## 6.2 Sales by category

The delivered-order category chart shows a fairly distributed mix. Home is the largest visible share at approximately 18.4%, followed closely by Grocery at 18.3% and Sports at 17.8%. Fashion contributes about 17.1%, Beauty about 17.0%, and Electronics about 11.4%. This suggests no single category dominates the displayed delivered sales, while Electronics is the smallest of the categories shown.

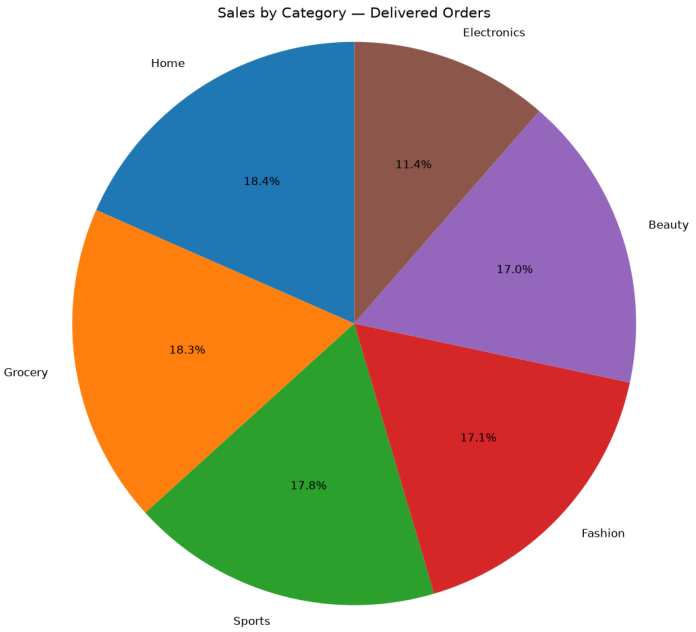



*Figure 5. Delivered-order sales by product category.*

## 6.3 Orders by status

The status chart shows Delivered: 271 orders (26.6%), Pending: 262 (25.7%), Returned: 243 (23.9%) and Cancelled: 242 (23.8%), for a total of 1,018 orders in this chart. Delivered orders are the largest group, but pending, returned and cancelled orders together make up approximately 73.4% of the displayed total. This is an operational signal worth investigating; returned and cancelled orders should be reviewed by category, payment method, state and shipping mode.

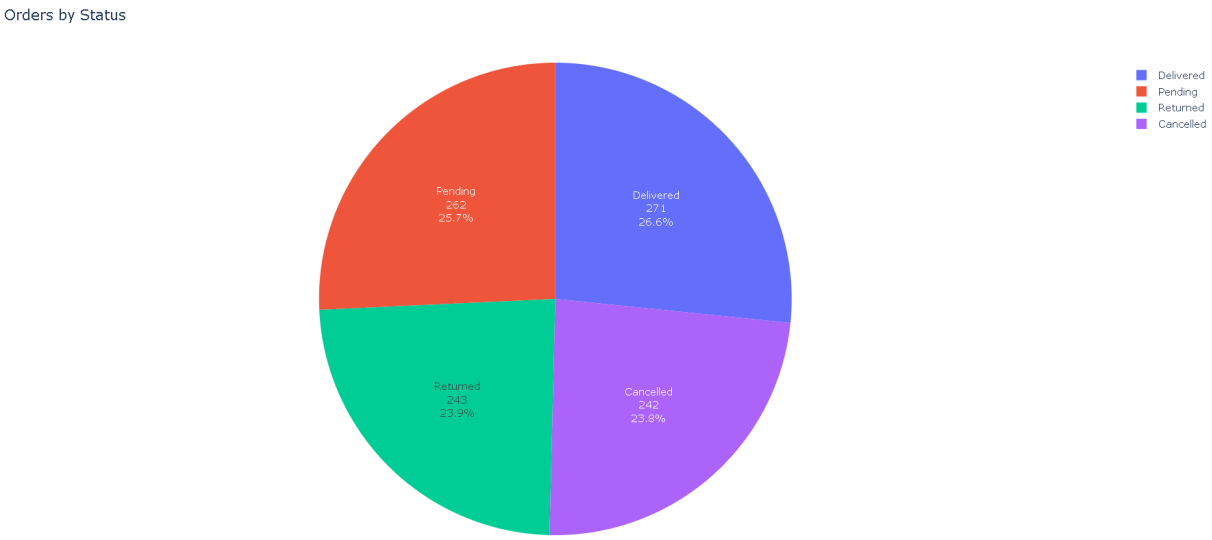



*Figure 6. Order-status distribution.*

## 6.4 Orders by payment mode

The payment chart displays Wallet 218 (21.4%), COD 213 (20.9%), Credit Card 204 (20.0%), Debit Card 204 (20.0%) and UPI 179 (17.6%). Wallet has the largest share in this chart, while UPI has the smallest. The distribution is relatively balanced, so payment-related conversion or failure rates should be compared before changing payment priorities.

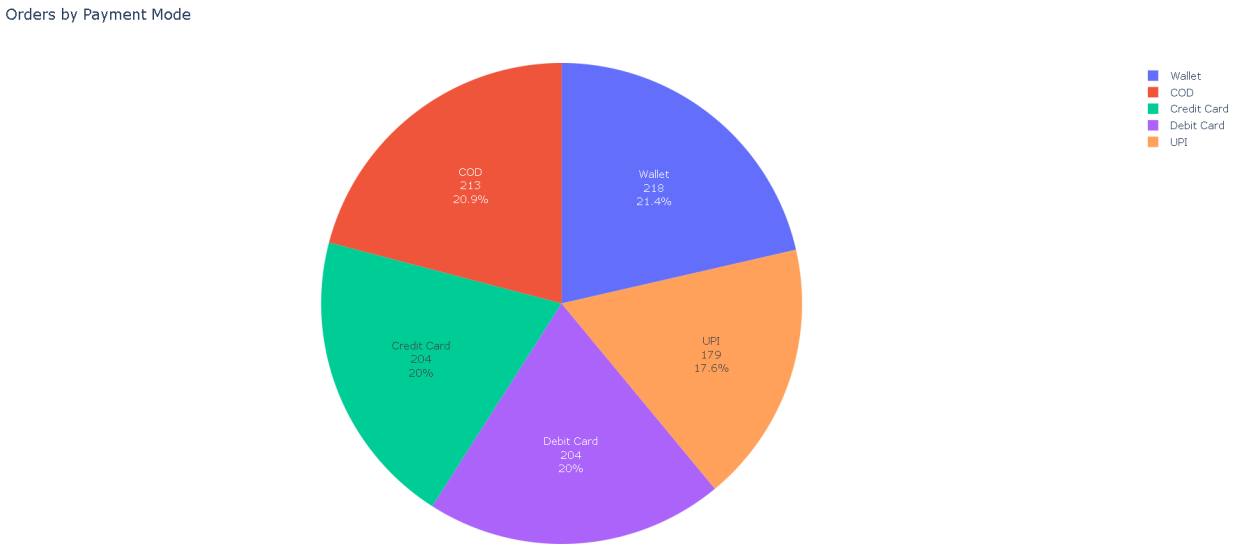



*Figure 7. Order distribution by payment mode.*

## 6.5 Orders by shipping mode

Express accounts for 347 orders (34.1%), Standard 342 (33.6%), and Same Day 329 (32.3%). The order mix is closely balanced across the three shipping modes. A useful next step is to compare delivery days, customer ratings, return rates and shipping costs across these modes, rather than assuming the most frequently selected option is automatically the best.

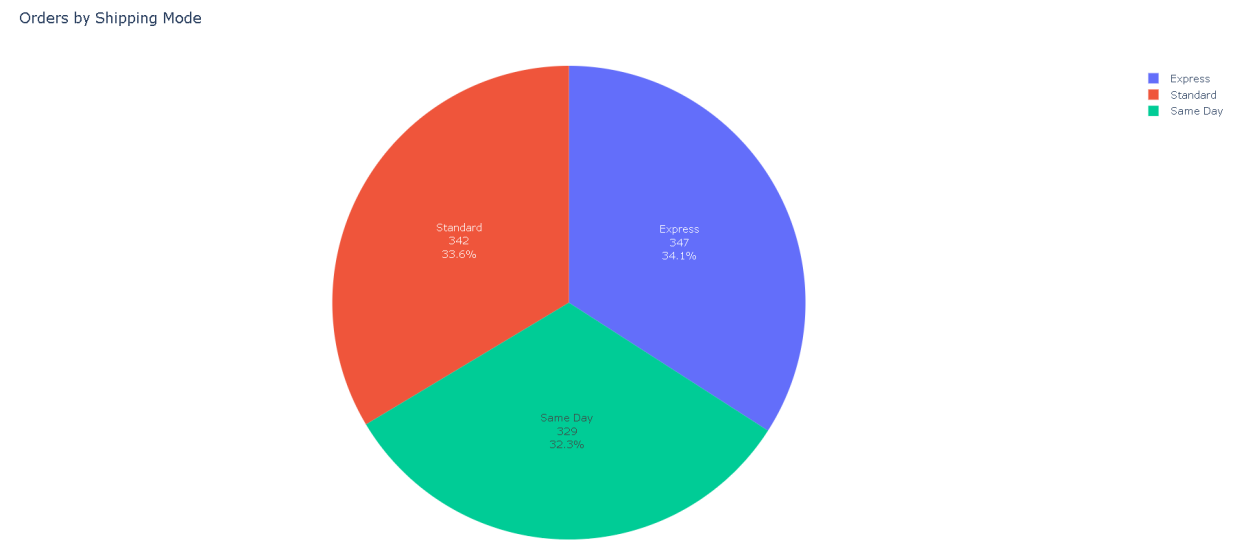



*Figure 8. Order distribution by shipping mode.*

## 6.6 Top customers by delivered sales

The supplied top-10 customer chart compares sales from delivered orders. Rahul Ali (C1222) is the largest visible slice at approximately 11.4%, followed by Omar Nguyen (C4949) at around 10.9% and Valentina Johnson (C7392) at around 10.7%. The chart is a relative-share view of the top ten, not necessarily their share of all business sales. Customer names and IDs should be handled carefully if this report is shared publicly.

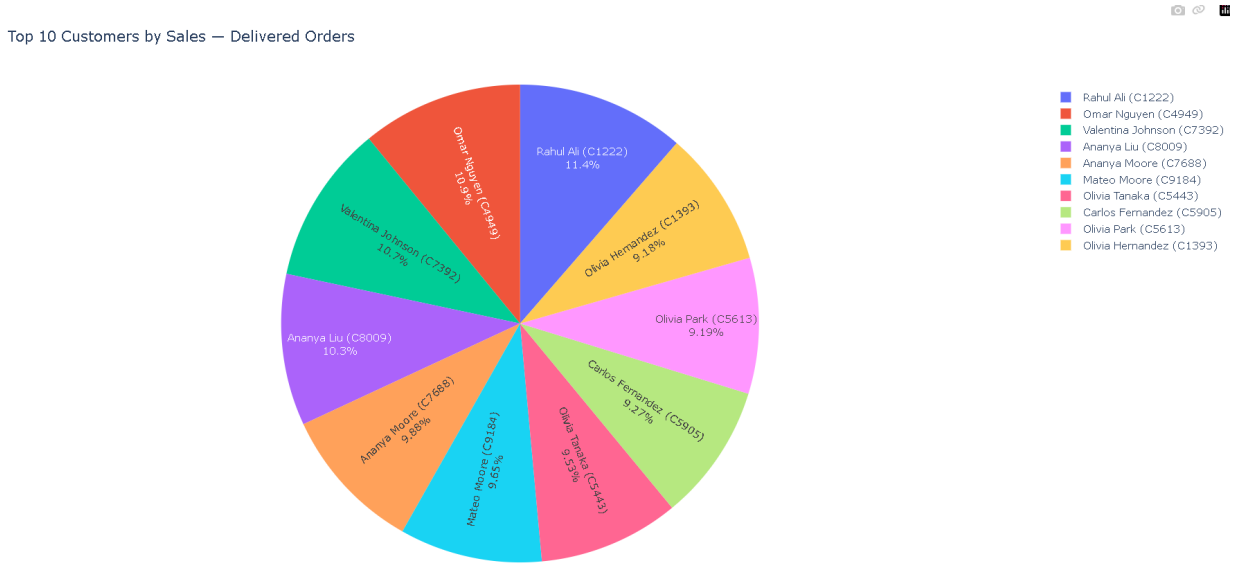



*Figure 9. Top 10 customers by sales for delivered orders.*

## 6.7 Sales by state

The state map indicates differences in delivered sales across US states. Texas appears among the highest-value states on the provided map, with California also showing a comparatively strong value. This chart can help prioritize regional inventory and marketing, but conclusions should be based on the underlying values table because color intensity alone does not show the exact state totals.

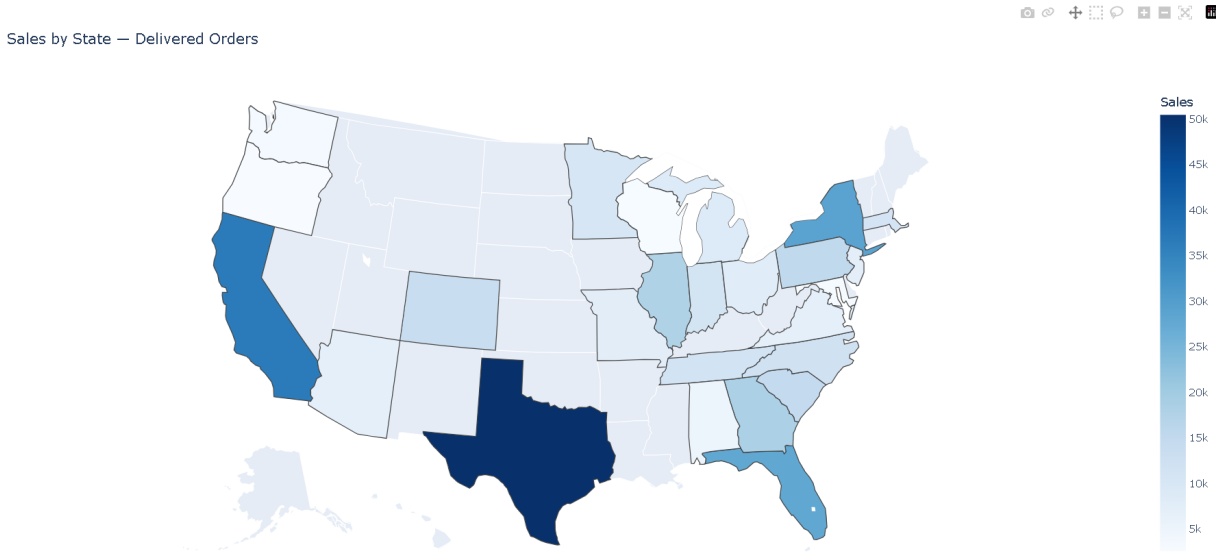



*Figure 10. Geographic distribution of delivered-order sales.*

# 7. Key Business Insights

**1. Order completion needs attention.** Only 26.6% of orders in the status chart are marked Delivered; pending, returned and cancelled orders together represent about 73.4%. Validate the status definitions and analyze why orders are not reaching Delivered.

**2. Category mix is broad.** Home, Grocery, Sports, Fashion and Beauty each contribute roughly 17–18% of delivered sales in the chart. Consider category-specific promotions and stock planning rather than focusing only on one category.

**3. Electronics has the smallest displayed category share.** At approximately 11.4%, Electronics is lower than the other shown categories. Check product availability, pricing, margin, returns and demand before deciding whether to increase promotion.

**4. Payment preferences are distributed.** Wallet, COD, credit card and debit card shares are close. Compare payment-mode conversion, failed transactions, return rates and fees to find opportunities.

**5. Shipping choices are nearly balanced.** Express, Standard and Same Day are within about two percentage points of one another. Assess profitability and service quality by shipping mode.

**6. Sales vary considerably by month.** The monthly chart contains clear peaks and troughs. Compare year-over-year values and check for incomplete periods before attributing changes to seasonality.

**7. Customer concentration should be monitored.** The top-customer chart ranks the strongest delivered-order customers. Track repeat purchases and customer retention, but avoid relying only on a top-ten chart.

**8. Regional sales can guide planning.** The state map suggests higher sales in some states, especially Texas and California. Validate with exact totals and consider order volume and profit alongside sales.

# 8. Recommendations

• Create a weekly order-status review for Pending, Returned and Cancelled orders; identify the top causes and assign corrective actions.

• Add a status funnel showing order count and sales at each stage, with a consistent date range and filter context.

• Compare category performance using sales, profit, order count, return rate and average rating together.

• Track month-over-month and year-over-year sales, and mark incomplete months clearly.

• Evaluate shipping modes using average delivery days, delivery delay rate, customer rating, shipping cost and return rate.

• Segment customers into new, repeat and high-value groups; build retention offers based on purchase behavior.

• Add filters for Year, Month, Category, State, Payment Mode, Shipping Mode and Order Status.

• Validate KPI definitions: distinguish total orders from unique customers and rename “Average Orders” if the measure is actually average order value.

# 9. Suggested Dashboard Layout

A professional interactive dashboard can be arranged into three sections:

**Top row — KPI cards:** Total Sales, Total Profit, Total Orders, Unique Customers, Average Order Value and Delivery/Return Rate.

**Middle row — performance:** Monthly sales and profit trend, sales by category, sales by state and top products.

**Bottom row — operations:** Order status, payment mode, shipping mode, delivery time and top customers.

Recommended global filters: Year, Month, Category, State and Order Status. Ensure every chart reacts consistently to the same filters. Keep colors consistent across visuals and show units/currency on all financial measures.

# 10. Tools and Skills Demonstrated

| Tool / Area | Use in this project |
|---|---|
| Microsoft Excel | Raw and cleaned worksheets, column checks, derived fields and data preparation. |
| Python (Pandas) | Loading the cleaned file, grouping and summarizing data, validating metrics and producing charts. |
| Matplotlib / Plotly | Monthly trend, category, status, payment, shipping and customer visualizations. |
| Tableau / BI dashboarding | Interactive presentation of KPIs, filters, category trends and geographic sales patterns; use the final tool actually implemented. |
| Business analysis | Interpretation of patterns, operational risks and recommendations. |

# 11. Limitations and Validation Notes

• The report is based on screenshots, not a direct recalculation from the workbook data. Numeric findings are transcribed from visible chart labels or estimated chart percentages.

• The top-level KPI dashboard and delivered-only charts may use different status filters; confirm the filter context before reconciling totals.

• The meaning of “Average Orders” is unclear from the screenshot. Confirm whether it means average order value, average quantity or another measure.

• The state visualization uses US states and city names. Ensure the report description matches the intended dataset context.

• The dataset appears to contain example-style customer and transaction values. Confirm data provenance and avoid presenting generated/sample data as actual company performance.

• The report does not claim that all cleaning checks were completed; document the exact actions performed in the final project README.

# 12. Conclusion

This e-commerce analytics project brings together data preparation, KPI reporting, trend analysis and operational visualizations. The charts suggest that sales vary by month, category contributions are broadly distributed, payment and shipping preferences are balanced, and order statuses deserve further investigation. The strongest next step is to validate the calculations against the cleaned workbook, make the KPI definitions explicit, and use the dashboard to monitor delivery outcomes, profitability and customer retention over time.

*End of Report*